

Project ATLAS — Day 4 Session Log: Investigation & Containment

Step-by-step record of work performed, with rationale, for SC200 study and SOC Analyst portfolio purposes

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Overview

Today's session executed **Day 4 (Investigation & Containment)** of Project ATLAS, transitioning from adversary simulation to analyst response. Working from the Microsoft Defender unified portal and Entra admin centre, the session covered incident triage, endpoint containment, risky user remediation, and proactive threat hunting — including an important telemetry gap discovery.

The session covered four areas:

1. Incident triage — triaging and classifying the live security incident generated by Day 3 attacks
 2. Endpoint containment — manually isolating DESKTOP-TRF9U79 via Windows Firewall
 3. Risky user remediation — confirming SOC Tor Test as compromised in Entra ID Protection
 4. Threat hunting — KQL investigation of endpoint telemetry pipelines, revealing a coverage gap
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Part A — Incident Triage

What was found

Navigating to **security.microsoft.com** → **Incidents & alerts** → **Incidents** revealed one live incident in the queue: **ID 2: "Initial access incident involving one user"**, severity High, status Active. The incident had 8 of 9 alerts active, sourced from Entra ID Protection (AAD Identity Protection), spanning activity from 3 July 2026 23:14 to 4 July 2026 00:07.

The incident graph linked the user entity **ivanyamoahbaoakye** to two distinct Tor exit node IP addresses. The Alerts tab listed six correlated detections: four Anonymous IP address alerts and two Anomalous Token alerts — all auto-correlated by Microsoft Defender into a single incident.

The right-side triage panel displayed an AI-generated technique assessment: **"Technique Profile: Using token theft and token replay for unauthorized access"** and Threat Overview: **"Cloud identity abuse"** — consistent with the Tor-based anonymous sign-in executed in Day 3.

Triage action taken

Opened the Manage incident panel and set **Classification: True positive — Compromised account**. Confirmed and saved. The incident header updated to show “**True alert**” status. The Assets (1) tab confirmed the single impacted asset: IVAN YAMOAHA BAOAKYE. The Summary tab showed: 1 impacted user, 2 IP addresses, 1 cloud logon session, first activity 3 Jul 2026 23:14:11.

Notable observation

Only one incident appeared in the queue, sourced entirely from Entra ID Protection (cloud identity telemetry). The Day 2 custom KQL analytics rules (brute force / encoded PowerShell) did not generate incidents — consistent with the endpoint telemetry gap discovered during the Day 5 hunting phase (see Part D). The Entra ID Protection pipeline was fully operational; endpoint telemetry was not.

Why this matters

Incident triage — correctly classifying a true positive vs. false positive, reading the incident graph, and documenting analyst reasoning — is the core daily function of a Tier 1 SOC analyst. The AI-generated technique profile demonstrates how Microsoft Defender’s built-in ML augments analyst triage speed without replacing analyst judgment. Directly reflects “lead and support incident response activities, including investigations” in the job description.

Part B — Endpoint Containment

Context: why manual containment instead of EDR isolation

Microsoft Defender for Endpoint’s “Isolate device” one-click action is unavailable because DESKTOP-TRF9U79 is onboarded via Azure Arc + AMA rather than the MDE sensor (a deliberate architecture decision from Day 1, chosen to avoid \$15/server/month Defender for Servers billing). The equivalent manual action — a host-based Windows Firewall block-all policy — achieves the same containment outcome and is what a Tier 1 analyst would do for an unmanaged or unenrolled host.

What was done

On DESKTOP-TRF9U79, opened an elevated PowerShell terminal (Terminal (Admin) via right-click Start) and ran:

```
netsh advfirewall set allprofiles firewallpolicy blockinbound,blockoutbound
```

Verified containment state with:

```
netsh advfirewall show allprofiles
```

Output confirmed **Firewall Policy: BlockInbound,BlockOutbound** across all profiles (Domain and Public). The endpoint was effectively isolated — all inbound and outbound traffic blocked at the host level.

Firewall was restored immediately after confirming the contained state, to preserve the Sysmon/AMA telemetry pipeline for ongoing monitoring:

```
netsh advfirewall reset
```

Why this matters

Endpoint containment is the first responder action in any compromise scenario. The ability to contain a host without EDR — using native OS controls — demonstrates depth beyond tool dependency, directly relevant to “proven experience supporting incident response in an enterprise environment, including investigation, containment, and remediation.”

Part C — Risky User Remediation

What was done

Navigated to **entra.microsoft.com** → **ID Protection** → **Risky users**. The page showed 2 risky users with a bar chart showing a spike of 2 new risky users on 4 July 2026. **SOC Tor Test** (soc■tor■test@VANPASSSC200.onmicrosoft.com) appeared in the list with Risk state: At risk, Risk level: High, last updated 5 July 2026.

Selected the SOC Tor Test user and clicked **Confirm user(s) compromised** in the toolbar. A green confirmation toast appeared: **“Initiated user compromised confirmation — Successfully initiated the user compromised confirmation process for user SOC Tor Test. It might take several minutes to complete the flow.”**

Why confirm compromised rather than dismiss

“Confirm compromised” is the correct SOC action for a verified true positive: it escalates the user’s risk state, triggers any configured automated response policies (e.g. force password reset, block sign in), and feeds back into Microsoft’s risk model to improve future detections. “Dismiss user risk” is for false positives only. This distinction is tested in SC■200 and is a real operational difference that affects downstream automation.

Why this matters

Risky user remediation in Entra ID Protection is a cloud native SOC response action with no on premises equivalent. Demonstrating the correct confirm vs. dismiss decision — and understanding why it matters for automated policy triggers — reflects “improve HOOPP’s cloud security posture” and the identity/access management depth called out in the job description.

Part D — Threat Hunting: Telemetry Gap Discovery

Hunting approach

Using **Microsoft Defender Advanced Hunting** (connected to workspace law■soc■atlas), ran two KQL queries to validate endpoint telemetry coverage before relying on it for detection confidence.

Finding 1: SecurityEvent table — 0 records

```
SecurityEvent | summarize count()
```

Result: **count_ = 0**. The SecurityEvent table exists (schema deployed by the Day 2 Windows Security Events via AMA DCR) but contains no ingested records. The Windows Security Events DCR was created and validated during Day 2 (schema confirmed, deployment succeeded) but event data never flowed. Likely cause: AMA agent heartbeat issue or DCR data stream misconfiguration. This means Event ID 4625 (failed logon) telemetry from the Hydra brute force attack in Day 3 was not captured in Sentinel.

Finding 2: Sysmon Event table — 0 records

```
Event | where Source == "Microsoft-Windows-Sysmon" | summarize count()
```

Result: **count_ = 0**. The Sysmon telemetry pipeline (configured in Day 1 via dcr soc atlas sysmon DCR) also has 0 records. This means the encoded PowerShell execution (Sysmon Event ID 1) from Day 3 was not captured. The Day 2 custom detection rules (ATLAS — Brute force failed logon spike; ATLAS — Encoded PowerShell execution) would not have fired even if the attack traffic matched their logic — because there was no data for them to query against.

Compensating control confirmed

Entra ID Protection telemetry (cloud identity layer) remained fully operational throughout — the Anonymous IP address and Anomalous Token detections fired correctly, the incident was created, and the risky user was flagged. This confirms that the attack was detected through the cloud IAM pipeline even while endpoint visibility was absent — a realistic defence in depth scenario where a single telemetry gap does not result in a complete blind spot.

Why this matters

Discovering that two telemetry pipelines are silent — not through tool alerts, but through proactive KQL hunting — is exactly the “detection gap” finding that justifies a threat hunting programme in a real SOC. A production analyst would escalate this to the SIEM engineering team with the query output as evidence, request an AMA agent health check, and flag which detection rules are currently non functional pending the fix. This directly reflects “improve detection, response, and alerting capabilities across security tools and platforms.”

Status Snapshot

Part	Action	Outcome
A — Incident triage	Classified ID 2 as True positive — Compromised account	Complete
B — Endpoint containment	BlockInbound,BlockOutbound via netsh on DESKTOP-TRF9U79	Complete — firewall state confirmed and restored
C — Risky user remediation	Confirm compromised: SOC Tor Test	Complete — confirmation initiated
D — Threat hunting	KQL queries: SecurityEvent + Sysmon Event count	Complete — telemetry gap discovered (both tables = 0 records)
E — Conditional Access	Report-only block policy on sign-in risk Medium/High	Planned — to be created in Entra Conditional Access post-session

Next session — Day 5: Hunting, Documentation & GitHub

1. Investigate and resolve the AMA agent / DCR telemetry gap (optional remediation for completeness)
2. Complete the Conditional Access policy (Report only mode) in Entra
3. Write the formal Project ATLAS Incident Report (final portfolio deliverable)
4. Push all documentation, session logs, and KQL queries to a public GitHub repository